EXECUTIVE SUMMARY

What The Backtest Is Actually Saying

This is a short decision memo, not the full report. It separates the investable core from the eye-catching but hard-to-survive ideas.

Generated 2026-05-11 06:22 UTC | Current regime: Fragile Late-Cycle Watch | Quadrant: Growth Down / Inflation Down | History tested: 2006-2026.

BOTTOM LINE

Keep The Unlevered Reserve Rule

тАв Best unlevered growth engine: Quality Rotation at about +5.7% CAGR with -53.0% max drawdown.
тАв Best steadier live profile: Reserve Rule at about +6.0% CAGR with -47.7% max drawdown.
тАв The 2000% number is total return in the leveraged ML path, not an annual rate.
тАв SPY x3 shows why leverage alone is dangerous: +8.7% CAGR with -92.4% max drawdown.

PRINCIPLE 1

Protect The Downside First

The reserve rule earns less than the fastest strategy, but it keeps the path smoother. That matters because strategies only work if you can hold them.

PRINCIPLE 2

Borrow Only After You Earn The Right

Leverage is not alpha. It amplifies whatever quality already exists. Here it raises reward and cuts survivability.

PRINCIPLE 3

Separate Prediction From Proof

The model signal is one thing. The proof is the untouched holdout and the walk-forward history. They are separate on purpose.

MEDIUM RISK / LOWER EFFICIENCY

Reserve Rule

This earns a return, but the drawdown path is harsher than the reward profile deserves.
CAGR +6.0% | Max DD -47.7%

MEDIUM RISK / LOWER EFFICIENCY

Quality Rotation

This earns a return, but the drawdown path is harsher than the reward profile deserves.
CAGR +5.7% | Max DD -53.0%

HIGH RISK / HIGH REWARD

Reserve x3

Reward can be large, but the path is violent and hard to hold in live capital.
CAGR +6.7% | Max DD -69.1%

EXTREME RISK / WEAK REWARD QUALITY

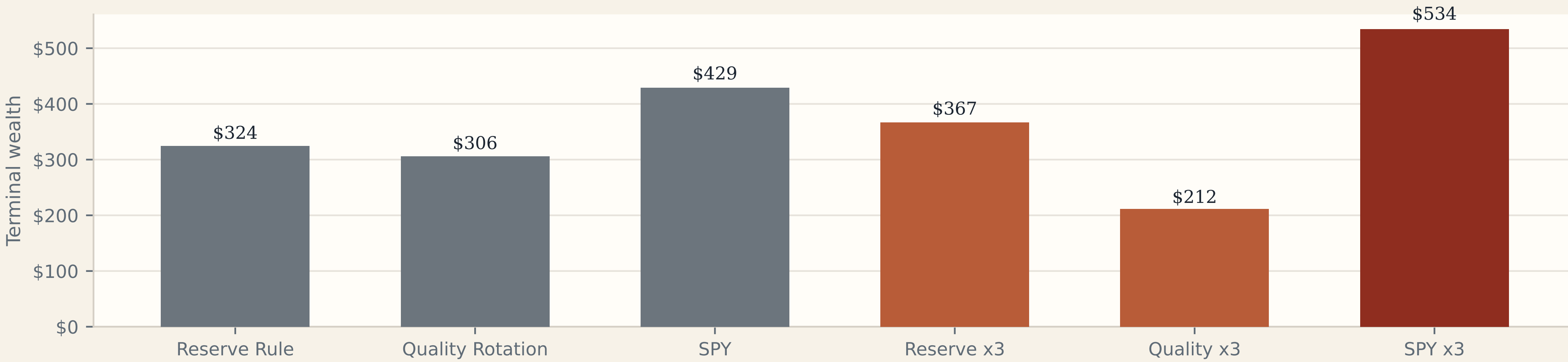
SPY x3

Borrowing magnifies the pain faster than it improves the compounding path.
CAGR +8.7% | Max DD -92.4%

Risk / Reward Map

Left is better. Up is better. Full 2006-2026 walk-forward history only.

Terminal Wealth From $100



This is why the leveraged rows show 2000%+ total return while still carrying severe path risk.

RISK LADDER

High Reward Is Not The Same As High Quality

Strategy	Bucket	Hist CAGI	Hist DD	Holdout CAG	Holdout DD
Reserve Rule	Medium risk / lower efficier	+6.0%	-47.7%	+7.5%	-0.9%
Quality Rotat	Medium risk / lower efficier	+5.7%	-53.0%	+11.0%	-1.4%
SPY	Medium risk / lower efficier	+7.5%	-49.3%	+11.0%	-2.3%
Reserve x3	High risk / high reward	+6.7%	-69.1%	+8.1%	-1.0%
Quality x3	High risk / high reward	+3.8%	-93.9%	+29.2%	-4.4%
SPY x3	Extreme risk / weak reward	+8.7%	-92.4%	+29.9%	-7.2%

WHY THIS IS NOT FORWARD-LOOKING

Controls Against Bias

тАв Features are lagged 1 bar before training and prediction.
тАв Each fold uses a 5-bar purge and 5-bar embargo.
тАв The final holdout starts on 2025-01-01 and is never used for model selection.
тАв Signals execute one bar later and hold 5 bars.
тАв Macro regime labels explain results but do not generate the historical trades.

DECISION

What To Stick To

тАв Core portfolio: unlevered reserve cash rule.
тАв Growth sleeve: unlevered quality rotation if you can tolerate the deeper drawdown path.